Poultry Shop Inventory Management System

Mini Capstone, Checkpoint 1 (Days 1–2)

1. Project title and client

Title: PoultryStock: Inventory Management System for a Poultry Shop

Client: A small neighborhood poultry shop (dressed chicken, eggs, live birds, feeds, and supplements) with one owner and two to three staff members.

2. Problem statement

The shop tracks stock in a notebook and in the owner's memory. Because poultry products spoil quickly, this causes real losses:

- The owner cannot tell at a glance how much stock is left, so items run out during busy days or pile up and spoil.
- Spoilage, breakage (eggs), and mortality (live birds) are not recorded, so losses are invisible.
- Restock orders to suppliers are made by text message and are not tracked, so nobody knows what was ordered, approved, or delivered.
- Staff can change records freely, and the owner has no history of who changed what.

PoultryStock gives the shop one place to record products, stock movements, and supplier orders, with clear low-stock alerts and an approval step for restocking.

3. User roles

Role	What they can do
Admin (Owner)	Everything: manage categories, products, and suppliers; delete records; approve or reject purchase orders; view full dashboard and all stock movements; manage staff accounts.
Staff	Register and log in; view products and suppliers; record stock movements (stock-out for sales, spoilage, adjustments); create purchase order requests; mark approved orders as received; edit only their own pending requests.

4. Main entities (besides users)

1. **categories**: groups such as Dressed Chicken, Eggs, Live Birds, Feeds, Supplements.
2. **suppliers**: farms and distributors the shop buys from.
3. **products**: each item sold, linked to a category and a default supplier.

4. **stock_movements**: every stock change (in, out, spoilage, adjustment), linked to a product and the user who recorded it.
5. **purchase_orders**: restock orders to a supplier, linked to the staff who requested and the admin who approved.
6. **purchase_order_items**: the products and quantities inside each order (makes products and purchase orders many-to-many).

5. ERD (table design)

Table	Columns
users	id, name, email, password, role (admin/staff), timestamps
categories	id, name, description
suppliers	id, name, contact_person, phone, address
products	id, category_id (FK), supplier_id (FK), name, sku, unit (kg/piece/tray/sack), price, stock_qty, reorder_level, timestamps
stock_movements	id, product_id (FK), user_id (FK), type (in/out/spoilage/adjustment), quantity, reason, timestamps
purchase_orders	id, supplier_id (FK), requested_by (FK users), approved_by (FK users, nullable), status (pending/approved/received/rejected), notes, timestamps
purchase_order_items	id, purchase_order_id (FK), product_id (FK), quantity, unit_cost

Relationships: category hasMany products; supplier hasMany products; supplier hasMany purchase_orders; product hasMany stock_movements; user hasMany stock_movements; purchase_order hasMany purchase_order_items; product belongsToMany purchase_orders (through items); user hasMany purchase_orders (as requester).

The diagram image is in the companion ERD and wireframes page.

6. Planned features mapped to every requirement

#	Required feature	Planned in PoultryStock
1	Authentication (Sanctum)	Register, login, logout, get current user; token saved in React and sent as Bearer header.

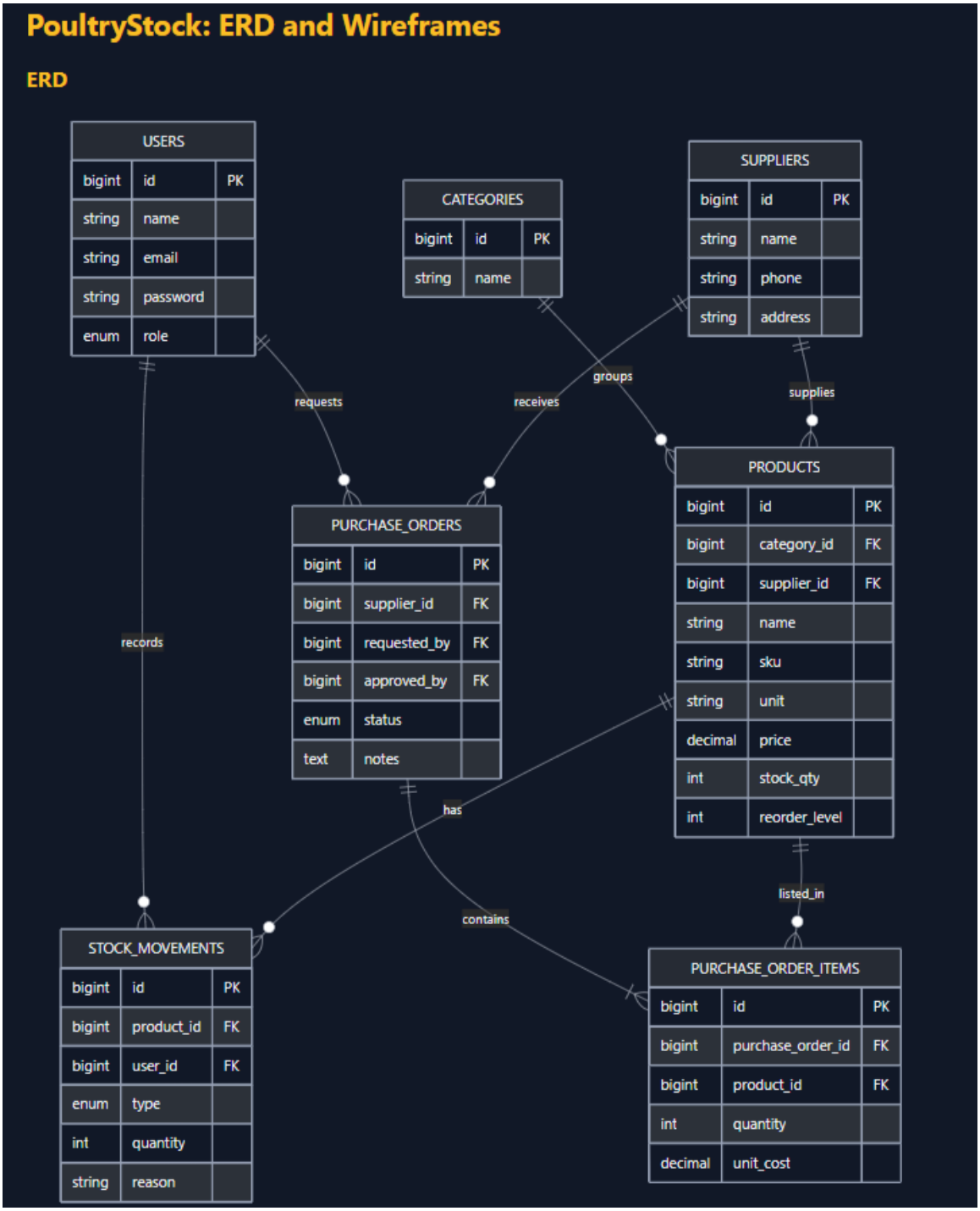
#	Required feature	Planned in PoultryStock
2	Role-based access	Admin and staff roles. Admin-only pages (Suppliers management, Users, Approvals) hidden in React and blocked in Laravel by middleware/policy.
3	Full CRUD, related resources	Products (belongs to category and supplier), Suppliers, Categories, and Purchase Orders with items.
4	Search and filter	Products searched by name or SKU, filtered by category and low-stock status; purchase orders filtered by status.
5	Pagination	Laravel paginate() on products, stock movements, and purchase orders, shown with page controls in React.
6	Dashboard (3+ figures)	Total products, low-stock items, pending purchase orders, total stock value, movements this week.
7	Validation on both sides	React form checks plus Laravel Form Requests (price > 0, quantity is a positive integer, unique SKU) with errors shown under fields.
8	User feedback	Loading spinners, success and error toasts, and a confirm modal before deleting.
9	Status workflow	Purchase order: Pending → Approved → Received (or Rejected). Staff creates; admin approves; staff marks received, which automatically adds stock and logs a movement.

Bonus candidates (later): dashboard charts with Recharts, Laravel Feature tests, CSV export of stock report.

7. Key pages for wireframes

Login/Register, Dashboard, Products list (search, filter, pagination), Product form, Stock movement form, Purchase orders list and detail (approval buttons for admin).

8. ERD diagram



9. Wireframes

